SAMPLE

Worlds Beyond Earth: Mars

The Red Planet of Ice and Dust

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Free sample: Introduction & Chapter 1
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Introduction

Mars is the world humanity knows best after its own, the red neighbor whose every era of watching has rewritten the previous one, the planet having been a god, then a coordinate, then a canal-riddled dream, then a cratered disappointment, and finally a place: the most mapped, most visited, and most argued-about world in the solar system, the argument being the book's whole engine and the engine being worth a full account in the series' manner.

The book tells Mars' story in the campaign's usual order: the welcome and the size arithmetic, the imagined landing and the thin air's honesty, the atmosphere's chemistry and the weather's calendar, the giants' geology and the secrets cabinet, the doomed moons and the canal era's strange biography, the spacecraft saga and the survival question, the life question and the frontier's files, the fact shelf and the closing lap the series grants every world it covers, the whole account being the red planet's biography for the resident of the blue one.

A calibration note, offered once: Mars is the two ledgers' own best demonstration in the whole series, because no world has been so thoroughly measured and so thoroughly planned at once. The measured: the surface having been photographed at every scale, the rovers having driven the actual ground, the quakes having been recorded, the ice having been mapped from orbit. The planned: the first crewed crossing being unfunded and unscheduled, the sample-return chain being half-built, the settlement visions being the second ledger's own residents, the reader carrying the settled and the open in the campaign's two hands, the two hands being the only posture that survives this subject's history.

The calibration deserves one companion note too, because Mars' history is the cautionary half of the method. The canal era having taught the field its hardest lesson: a whole generation of observers, instruments excellent for their day, having watched the same planet and seen what expectation prepared, the preparation having been irrigation grids on a desert world, the grids having been a misreading of blur at the eye's own limits, the misreading having lasted decades and inspired a literature, and the lesson being the campaign's own foundation: the eye is an instrument with a past, the past shapes what the instrument reports, and the only cure is the ledger discipline that names which readings are measurements and which are wishes, the naming being this book's constant practice.

The companion deserves a closing paragraph as well, because the era after the canals owes the reader its honest shape. The spacecraft having arrived and found craters where the dream had lakes, the disappointment being real and the rebound being better: the orbiters having found river valleys anyway, the valleys being ancient and real, the rovers having found lakebeds and river deltas in the actual rocks, the ancient wetness being therefore the modern consensus and the consensus being receipted by chemistry, the chemistry being the reversal's whole beauty: the world proved wetter than the dream imagined and colder than the dream allowed, both at once, the both-at-once being the kind of verdict only instruments can deliver, the delivery being why the whole account trusts the measured first.

The reader's practical stake also earns a paragraph of its own, because the account ends with a usable skill rather than a shelf ornament. The headlines about the red world arriving in a constant stream, the stream being the most headline-covered world off Earth, the coverage being the exact habitat of the ledgers' merger, the merger being what the reader is being armed against: every claim in every later headline being gradeable in the minute the account promised, the grading being the book's real product, the product being portable and the portability being the point.

The sibling volumes deserve one naming too, because the series' own map helps the reader place this one. The Earth volume having covered the home world's own machinery, the Venus and Mercury volumes having covered the neighbors inside this orbit, the Moon volume having covered the companion, the Pluto volume having covered the far cold archive: this account being the one for the world the species watches most, the placement being honest rather than flattering and the distinction being earned by traffic rather than beauty.

The book's own promise deserves the final opening line too, because the campaign signs its contracts up front. No claim will arrive unlabeled, no open file will be dressed as a discovery, no schedule will be quoted without its ledger, the three being the account's whole constitution, the constitution being short enough to memorize and strict enough to matter, the mattering being measured by how many headlines the reader will grade correctly afterward, the grading being the only tuition this book charges.

One framing fact opens the account, the single thing that explains the planet's hold on the imagination. Mars is the

only world whose surface a reader can effectively stand on today, in the sense that its landscape is public: mapped from orbit at human scale, walked by robots whose tracks are visible from space, and rendered so completely that a reader can browse a canyon wall as easily as a city street. No other planet offers that intimacy, and the chapters that follow use it, moving from the standing-on-it questions of the early chapters to the search-for-life questions of the later ones, on the strength of the most thoroughly known landscape beyond the Earth.

Chapter 01: Welcome to the Planet

Every account begins with the subject in the sky, and Mars' sky story is the oldest astronomy there is: a bright red wanderer noticed by every civilization that kept records, the noticing being the welcome's own prehistory and the prehistory being worth its full telling.

The naked-eye view deserves the first paragraph, because Mars earned its name before anyone knew what it was. The planet being one of the five wanderers the ancient watchers tracked against the fixed stars, the redness being its own signature even to the unaided eye, the redness earning war-god names in more than one tradition, the naming being humanity's first Mars science: accurate in color and wrong in everything else, the everything else waiting for the telescope, the telescope waiting for glass good enough to survive its own atmosphere, the atmosphere being the whole problem the early watchers shared.

The brightness warrants a separate paragraph, because Mars is the most variable of the bright planets. The orbit being distinctly oval, the planet's distance swinging wide between apparitions, the opposition years being bright and the off years being modest, the brightest apparitions having the planet outshine every star and rival the giants, the giants being steady and Mars being moody, the moodiness being the welcome's own honest note: the neighbor is close enough to change with geometry, the geometry being the reader's own schedule, the schedule being the opposition cycle the sky watchers still keep.

The opposition cycle deserves its paragraph as well, because the geometry explains the whole calendar of Mars watching. The Earth catching the slower planet every twenty-six months, the catch being the opposition, the opposition being the close pass, the close pass being when the planet looks largest and the telescopes work hardest, the cycle also being the launch-window rhythm the space era inherited wholesale, the inheritance being the welcome's quiet modernity: the same geometry that scheduled the watchers' nights now schedules the flight directors' windows, the windows being the sky's own clockwork and the clockwork being unbroken since the first watchers.

The place in the system gets a second paragraph here, because the address sets the character. The fourth orbit, the orbit beyond the Earth's, the sunward edge of the frost line's neighborhood, the distance being about one and a half times the Earth's own, the light arriving dimmer and redder, the year being nearly two Earth years and the day being nearly an Earth day, the day-length being the welcome's own uncanny gift: the visitor's internal clock nearly matching home's, the matching being an accident of rotation and the accident being one of the few things Mars gives for free, the free things being worth counting on a world that gives so few.

The character deserves the closing paragraph, because the welcome should end with the summary the later chapters cash. The world being small, cold, thin-aired, rusty, and reachable, the reachability being the defining property: no other world offers both genuine otherness and genuine accessibility, the otherness being complete and the accessibility being real, the combination making Mars the permanent second destination of the species' attention, the attention being the welcome's whole justification and the justification being the book's own contract with the reader, the contract being honored in the chapters ahead.

The amateur era merits a paragraph to itself, because the welcome continues in the modern back garden. The planet being the one bright world whose surface details are genuinely visible in a modest telescope at opposition, the caps and the dark markings being findable by the patient observer with ordinary glass, the finding being the amateur community's own annual ritual, the ritual keeping the naked-eye era's continuity alive: the same opposition calendar the ancients tracked now cueing eyepiece nights on the same schedule, the schedule being the one thread that runs unbroken from the war-god names to the webcam era, the thread being the welcome chapter's own closing receipt.

The color deserves one more receipt too, because the name is honest chemistry. The red being iron oxide throughout the dust, the oxide being the surface's own pigment, the pigment being the one property the naked eye always measured correctly, the correctness being a small vindication for the ancient watchers: they named the planet for the color and the color has survived every later instrument's audit, the audit being the welcome chapter's

own closing gift to the first sky-keepers.

The brightness extremes deserve one more receipt too, because the moody neighbor swings harder than any other. The closest oppositions having the planet blaze at the sky's own first rank, the farthest apparitions having it dwindle to a modest star, the swing being the oval orbit's own geometry, the geometry being the same arithmetic the launch windows ride, the ride being the welcome chapter's last quiet connection: the sky's oldest calendar and the rocket era's manifest being one document, the document being open to any reader who counts months.

The planet's standing in the sky earns one more welcome note, because the reader will meet it personally. The steady ruddy light hanging in the evening months, brightening to outshine every star in its best seasons, has been the sky's most familiar wanderer for every generation that ever looked up, and the modern observer needs no equipment at all to find it: a planet that does not twinkle, drifting eastward night by night against the stars, doubling back in its famous loops each couple of years, the loop the ancient watchers could not explain and the book's middle chapters will. The welcome chapter's standing invitation is accordingly the simplest in the series: step outside in a good month, find the red light, and hold the thought that everything in this book is happening there tonight, at the speed of its own year and its own clock, under a butterscotch sky.